

3D Scene Reconstruction and Virtual Tour

Amna Raza Khan
24280055@lums.edu.pk

Verda Batool
24280065@lums.edu.pk

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Abstract

This project implements a complete Structure from Motion pipeline to reconstruct 3D scenes from 2D photographs and create an interactive virtual tour experience. The project starts with phase 1, i.e., feature detecting and matching using SIFT with Lowe’s ratio test. This is followed by a two-view reconstruction with the help of essential matrix estimation and pose disambiguation. This is then extended to multi-view reconstruction by employing incremental perspective-n-point pose estimation. This also employs Bundle Adjustment refinement to reduce the drift.

This is the basis of an interactive viewer using Three.js in which users can navigate between different camera viewpoints smoothly. The final implementation transforms the dataset of overlapping images into a 3D environment that is navigable, showing the practical applications of multi-view geometry principles.

Project **Repository:** <https://github.com/verdabatool/structure-from-motion-3D-reconstruction>

1 Introduction

The main challenges and opportunities in computer vision are to construct three-dimensional scenes from a two-dimensional image. To address this problem, we use Structure from Motion (SfM). SfM extracts geometric information from a dataset of images. This geometric information helps in the development of applications ranging from augmented reality, autonomous navigation, virtual tours and preserving cultural heritage. Many commercialized projects revolve around the concept of 3D scanning, such as Matterport, which is for real estate and construction. These projects have great potential as the product is immersive, as they make an explorable environment from ordinary images. These technologies enable users to navigate around digitally reproduced environments by transforming collections of passive photos into interactive experiences.

The project is based on three key phases: implementing a robust and efficient two-view reconstruction, which then extends to multi-view SfM with bundle adjustment, and the final

phase comprises a three.js viewer for interactive visualization. The key contributions of this report are:

1. Implementation of a complete SfM pipeline from feature matching to bundle-adjusted multi-view reconstruction, demonstrating practical application of epipolar geometry, triangulation, and Perspective-n-Point algorithms.
2. Development of a registration framework that transforms independent partial reconstructions—each with its own local coordinate systems—into a unified global representation suitable for interactive visualization.
3. Creation of an interactive virtual tour using Three.js that provides smooth navigation between camera viewpoints with animated transitions, incorporating both point cloud visualization and high-resolution source images.
4. A comprehensive analysis of the challenges encountered in the practical implementation of SfM, including robustness to feature matching, accumulation of drift in incremental reconstruction, and alignment of the coordinate system for fragmented scenes.

2 Methodology: Structure from Motion

The Structure from Motion (SfM) pipeline implemented in this project works as a framework to reconstruct 3D scenes from 2D images.

The 3 main steps are as follows:

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2.1 Phase 1: Two-View Foundation

The first stage is two-view reconstruction, i.e., to establish the fundamental geometric relationship between the cameras and the scenes

Camera Intrinsic Estimation: The camera’s internal parameters are taken from EXIF metadata using the 35mm equivalence principle. For an image with focal length f_{mm} and 35mm-equivalent focal length f_{35} , the sensor width is calculated as $w_{\text{sensor}} = 36 \times \frac{f_{\text{mm}}}{f_{35}}$. The focal length in pixels is then

$f_x = \frac{f_{\text{mm}}}{w_{\text{sensor}}} \times w_{\text{px}}$, with the principal point (c_x, c_y) in the center of the image. The camera intrinsic matrix $\mathbf{K}$ is defined as:

$$\mathbf{K} = \begin{pmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{pmatrix}$$

Feature Detection and Matching: Scale-Invariant Feature Transform (SIFT) is used to detect distinctive keypoints in image 1 and image 2. Each keypoint is described by a 128-dimensional vector that captures local gradient information. Feature matching uses a FLANN-based k-nearest neighbor search with Lowe’s ratio test (ratio = 0.75) to filter out vague matches. The ratio test is formulated as: $\frac{\text{distance}(\mathbf{d}_i, \mathbf{d}_{j1})}{\text{distance}(\mathbf{d}_i, \mathbf{d}_{j2})} < \text{ratio}$.

Essential Matrix Estimation: From normalized image coordinates $\mathbf{x}$ and $\mathbf{x}'$, the essential matrix $\mathbf{E}$ satisfies the epipolar constraint: $\mathbf{x}'^\top \mathbf{E} \mathbf{x} = 0$. We have used RANSAC with a 1-pixel threshold and $\mathbf{E}$ is estimated from at least 5 point correspondences, handling outlier matches robustly. The essential matrix shows how two cameras are related to each other. It captures the rotation and translation between the two cameras. Also, point correspondence means finding the same points in both images.

Pose Recovery: The essential matrix is decomposed into relative rotation $\mathbf{R}$ and translation $\mathbf{t}$ using singular value decomposition, followed by cheirality checking to select the physically valid pose. The SVD of the Essential Matrix is $\mathbf{E} = \mathbf{U}\Sigma\mathbf{V}^\top$.

Pose Disambiguation: The essential matrix is broken down into four possible combinations of $[\mathbf{R}|\mathbf{t}]$ camera poses. The correct pose is selected through *cheirality checking*; only one configuration places triangulated points in front of both cameras (positive depth). Mathematically, for a point $\mathbf{X}$ in world coordinates, we require $(\mathbf{R}_i \mathbf{X} + \mathbf{t}_i)_z > 0$ for both cameras $i = 1, 2$.

Triangulation: Triangulation is the process of determining the 3D coordinates of a point in space by intersecting lines of sight from two or more known camera positions. With known camera poses $\mathbf{P}_1 = [\mathbf{I}|\mathbf{0}]$ and $\mathbf{P}_2 = [\mathbf{R}|\mathbf{t}]$, 3D points are reconstructed via linear triangulation solving $\mathbf{x} \times (\mathbf{P}\mathbf{X}) = \mathbf{0}$ for each correspondence. This forms a linear system that is typically solved as:

$$\begin{pmatrix} [\mathbf{x}]_{\times} \mathbf{P}_1 \\ [\mathbf{x}']_{\times} \mathbf{P}_2 \end{pmatrix} \mathbf{X} = \mathbf{0}$$

The resulting sparse point cloud forms the initial 3D reconstruction.

2.2 Phase 2: Incremental SfM with Bundle Adjustment

The two-view reconstruction is extended to multiple images through incremental registration and global refinement.

Incremental Camera Registration: For each new image I_i , feature matches are established with previously registered

images. Existing 3D points visible in I_{i-1} provide 2D-3D correspondences for Perspective-n-Point (PnP) estimation. Using RANSAC PnP, the camera pose $[\mathbf{R}_i|\mathbf{t}_i]$ is estimated by minimizing reprojection error:

$$\min_{\mathbf{R}, \mathbf{t}} \sum_j \|\mathbf{x}_j - \pi(\mathbf{R}\mathbf{X}_j + \mathbf{t})\|^2$$

where π is the projection function.

Map Expansion: New triangulation occurs between I_i and all previous images using the estimated pose. Points are only added if they satisfy multiple criteria: positive depth in both views, sufficient triangulation angle ($\theta > 1^\circ$), and reprojection error below 3 pixels. The triangulation angle θ is calculated from the normalized viewing vectors $\mathbf{v}_1$ and $\mathbf{v}_2$:

$$\theta = \arccos\left(\frac{\mathbf{v}_1 \cdot \mathbf{v}_2}{\|\mathbf{v}_1\| \|\mathbf{v}_2\|}\right)$$

This conservative approach ensures reconstruction quality.

Bundle Adjustment Bundle adjustment optimizes all camera parameters $\mathbf{R}_i, \mathbf{t}_i, \mathbf{K}$ and 3D points $\mathbf{X}_j$ by minimizing the total reprojection error:

$$\min_{\mathbf{R}_i, \mathbf{t}_i, \mathbf{X}_j} \sum_{i=1}^m \sum_{j=1}^n v_{ij} \|\mathbf{x}_{ij} - \pi(\mathbf{K}[\mathbf{R}_i|\mathbf{t}_i]\mathbf{X}_j)\|^2$$

where v_{ij} is a visibility indicator. The optimization uses the Levenberg-Marquardt algorithm with sparse Jacobian structure, exploiting that each residual depends only on one camera and one point.

Implementation Details: Our implementation fixes the first two cameras to establish the coordinate frame and optimizes subsequent cameras. The optimization shows 8.2% improvement.

2.3 Phase 3: Virtual Tour Visualization

The reconstruction enables an interactive experience where users navigate between original camera viewpoints.

Camera Graph Construction: Registered cameras become nodes in a view graph, with edges weighted by shared visible 3D points. The weight w_{12} between two cameras is given by $w_{12} = |\text{Points}(\text{Camera}_1) \cap \text{Points}(\text{Camera}_2)|$. This graph enables intelligent navigation between logically connected viewpoints.

View Interpolation: Smooth transitions between cameras $[\mathbf{R}_1|\mathbf{t}_1]$ and $[\mathbf{R}_2|\mathbf{t}_2]$ use spherical linear interpolation (slerp) for rotation and linear interpolation for translation:

$$\begin{aligned} \mathbf{R}(t) &= \text{slerp}(\mathbf{R}_1, \mathbf{R}_2, t) \\ \mathbf{t}(t) &= (1-t)\mathbf{t}_1 + t\mathbf{t}_2 \end{aligned}$$

Simultaneously, images cross-fade using alpha blending: $I(t) = (1-t)I_1 + tI_2$.

2.4 Interactive Virtual Tour

The interactive virtual tour is the final stage of the Structure from Motion (SfM) pipeline, leveraging the accurately reconstructed sparse 3D scene and the precisely estimated camera poses to create an immersive, navigable experience.

2.4.1 1. Camera Graph Construction

The foundation of the tour’s navigability is the **Camera Graph**, where each successfully registered camera pose represents a node.

- **Nodes:** The camera poses $[\mathbf{R}_i | \mathbf{t}_i]$ and the corresponding original image I_i become the nodes in the graph.
- **Edges:** An edge connects two cameras if they share a significant number of visible 3D points. This criterion ensures that the connected views have sufficient **overlap** and a common visual context. The weight of the edge w_{12} between two cameras is typically proportional to the count of shared 3D points:

$$w_{12} = |\text{Points}(\text{Camera}_1) \cap \text{Points}(\text{Camera}_2)|$$

- **Function:** This graph allows the system to guide the user along logical, connected pathways in the reconstructed scene, rather than jumping arbitrarily between views.

2.4.2 2. Smooth View Transition

To prevent jarring jumps between discrete camera positions, the tour employs interpolation techniques for both the camera movement and the visual display:

- **Pose Interpolation:** When moving from camera 1 to camera 2, the intermediate poses $[\mathbf{R}(t) | \mathbf{t}(t)]$ are generated using interpolation over a time parameter $t \in [0, 1]$.
 - **Rotation:** **Spherical Linear Interpolation (Slerp)** is used for smooth rotation: $\mathbf{R}(t) = \text{slerp}(\mathbf{R}_1, \mathbf{R}_2, t)$. Slerp ensures the camera rotation maintains a constant angular velocity.
 - **Translation:** Standard **Linear Interpolation (Lerp)** is used for position: $\mathbf{t}(t) = (1 - t)\mathbf{t}_1 + t\mathbf{t}_2$.
- **Visual Blending:** The images are cross-faded using **alpha blending** to smooth the visual transition as the camera moves:

$$I(t) = (1 - t)I_1 + tI_2$$

This technique creates the illusion of moving through the static image set.

2.4.3 3. User Interaction and Visualization

The final interactive product allows the user to experience the 3D scene:

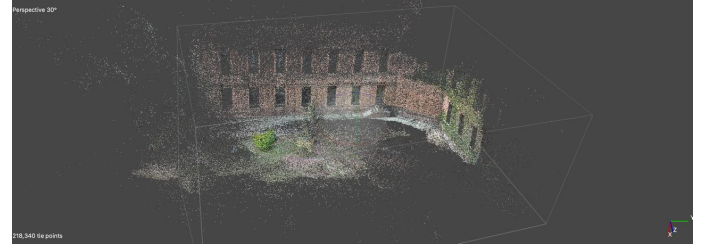


Figure 1: Screenshot of the point cloud with a small amount of drift .

- **Navigation:** Users can click on designated hotspots or points of interest to trigger a transition to a nearby, relevant viewpoint in the camera graph.
- **Visualization:** The sparse point cloud is often rendered alongside the current image view, providing context and showing the user which part of the 3D map they are currently observing. This helps visualize the extent and density of the reconstruction.

The interactive virtual tour effectively transforms the abstract geometric output of the SfM process into a tangible, easy-to-understand representation of the original scene.

3 Experimental Results

This section presents the results of the Structure from Motion (SfM) pipeline, demonstrating the quality of the 3D reconstruction before and after the global optimization step, known as Bundle Adjustment (BA). The performance analysis details the initial setup and the incremental registration process.

3.1 3D Reconstruction Quality: Pre- vs. Post-Bundle Adjustment

The most critical factor in multi-view reconstruction is the accuracy and cohesion of the sparse point cloud. **Bundle Adjustment** globally refines all camera poses and 3D points simultaneously, minimizing the overall reprojection error across all views.

Figure 3.1 compares the point cloud reconstruction at two key stages: immediately following incremental triangulation (Pre-BA) and after the non-linear optimization (Post-BA).

The **Post-Bundle Adjustment** views exhibit a visibly tighter and more geometrically coherent structure compared to the Pre-BA views, where noise and accumulated errors from the incremental step caused significant geometric drift

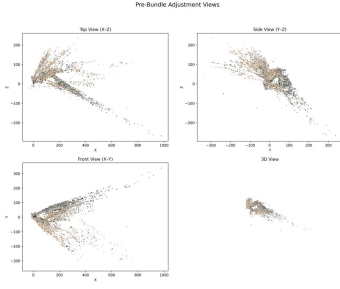


Figure 2: Pre-Bundle Adjustment Views

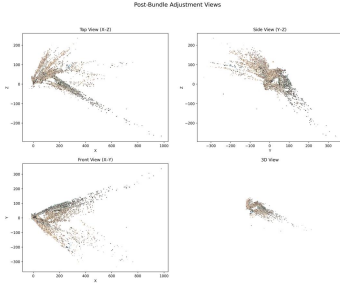


Figure 3: Post-Bundle Adjustment Views

and a less dense, scattered point cloud. Quantitatively, the Bundle Adjustment step yielded a **8.2%** improvement in the reconstruction’s accuracy (measured by the reduction in the final average reprojection error).

Comparison of the 3D Sparse Point Cloud before and after Bundle Adjustment (BA) optimization. Note the improved cohesion and reduced geometric “bending” or “drift” in the post-BA reconstruction, particularly noticeable in the Top View (X-Z) and Front View (X-Y) plots.

3.2 Quantitative Performance Analysis

The quantitative results track the feature detection, initialization, and incremental registration of the camera poses and 3D map points.

3.2.1 Initialization and Feature Extraction

The pipeline loaded 384 images, successfully extracting SIFT keypoints from all of them. The two-view initialization used images 0 and 1, establishing the initial coordinate system:

- **Average Keypoints per Image:** $\approx 14,000$ (Varies from $\approx 2,700$ to $\approx 26,600$).
- **Initial Matches (Images 0 $\leftrightarrow$ 1):** 3609 good matches.
- **Essential Matrix Inliers (RANSAC):** 3063 out of 3609.
- **Initial 3D Points:** 3062 points (after cheirality and filtering).

4 Discussion and Limitations

Feature Matching: We faced a problem while selecting the dataset, as blank walls and uniform areas are textureless surfaces. The inlier ratio was low, despite the presence of many initial matches. Furthermore, the Ratio test (0.75) was too conservative for some pairs of images, and the scale invariance was helpful, but it was not sufficient for large viewpoint changes.

Geometric Estimation: The essential matrix is sensitive to pure rotation and planar scenes (motion). Also, while implementing Cheirality, which assumes that the scene points are always in front of the camera, this is not true in every scenario. Additionally, there’s the problem of scale ambiguity in monocular reconstruction as it loses the original depth.

5 Conclusion

This project demonstrates a full pipeline for turning the 2D points into an interactive 3D world tour. The first stage consists of Feature matching using SIFT and Lowe’s ratio test. The t3ww view phase was done to generate the initial sparse cloud using essential matrix estimation, pose disambiguation via cheirality checking, and linear triangulation. Phase two, multi-view SFM had camera registration using Perspective-n-Point estimation and map expansion through triangulation implemented. Bundle adjustment provided measurement refinement, improving the RMS projection by 8.2/

This work offers a useful framework for comprehending the trade-offs between algorithmic complexity, computational efficiency, and reconstruction quality in 3D computer vision systems in addition to a pedagogical basis in multi-view geometry.